

Summary of Integration Worksheet

Compute each of the following. Use algebra and u -substitution when appropriate.

1. $\int \frac{\tan^{-1}(x)}{1+x^2} dx$

2. $\int_0^{\frac{\pi}{2}} \cos(x) \sin(\sin(x)) dx$

3. $\int \tan(x) \ln(\cos(x)) dx$

$$4. \int_0^{\frac{1}{4}} \frac{\sin^{-1}(2x)}{\sqrt{1-4x^2}} dx$$

$$5. \int \frac{t^3 - 4t^2 + 4t}{t^2 - 2t} dt$$

$$6. \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \tan^3(\theta) d\theta$$

$$7. \int \frac{x^2}{x^2+1} dx$$

$$8. \int_2^4 \frac{x^4-16}{x-4} dx .$$

$$9. \int x^3 \sqrt{x^2+1} dx$$

$$10. \int \frac{2e^z+2}{e^z+z} dz$$

$$11. \int_0^1 \left(\sqrt[4]{x} + 1 \right)^2 dx$$

$$12. \int \left[7 + \cot^2(x) \right] dx$$

$$13. \int_{-1}^0 \frac{x^2}{\sqrt{1-x}} dx$$

14. $\int \frac{x^5 + x^2}{1 + x^6} dx$

15. $\int \frac{3x^8}{x^3 + 4} dx$

16. $\int_{-1}^4 \frac{2x^2 + 7x^5}{x^4} dx$

17. $\int e^{5x} dx$

18. $\int x\sqrt{x^2-1} dx$

19. $\int_1^4 \frac{4x^3 - \sqrt{x}e^x - 3\sin(\sqrt{x})}{3\sqrt{x}} dx$